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Menstrual product use patterns in a large digital cohort in the United States: Variations by sociodemographic, health, and menstrual characteristics

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Abstract

Background: Using menstrual products is a part of managing menstrual bleeding. Products use may represent individual, social, and economic influences. A few studies on menstrual product use from specific regions in the United States (US) reported differences in use across demographic factors like age and race and ethnicity.

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Conflict of Interest Disclosures

C.L.C. and T.F.C. own Apple Inc. stock and are employed by Apple Inc. Other co-authors have no conflict of interest. There were no financial relationships with any organizations that might have an interest in the submitted work in the previous 3 years and no other relationships or activities that could appear to have influenced the submitted work.

Objectives: To evaluate menstrual product use patterns and how they vary by sociodemographic, health, and menstrual characteristics in a large US cohort.

Study Design: We used data from 11,455 participants across the US from the Apple Women's Health study, an ongoing mobile application-based US cohort, who provided consent, enrolled between 11/2019–1/2024, and responded to relevant surveys. Data included age, race and ethnicity, ancestry, gender identity, socioeconomic status (SES), gynecological conditions, menstrual characteristics, use of menstrual products (panty liners, sanitary napkins or pads, regular tampons, super-absorbent tampons, a combination of pads or tampons but not used together, internal cup menstrual collection device, period underwear, or other), and whether sleeping with a tampon among tampon users. We obtained the percentages and 95% confidence intervals (CIs) of each product use, overall and by several characteristics. We used Chi-square tests and 95% CIs to compare percentages across categorical characteristics.

Results: Overall, the prevalence of regular tampons use was 48%, followed by pads (47%), panty liners (43%), super-absorbent tampons (36%), period underwear (20%), cups (18%), a combination of pads or tampons but not used together (18%), other products (2%), and no product (4%). Among tampon users, 61% sleep with a tampon. Compared to individuals aged 30–39, those aged <20 were more likely to use pads (57% [95% CI: 53–61] vs. 45% [95% CI: 43–46]) and less likely to use cups (12% [95% CI: 10–15] vs 21% [95% CI: 19–22]) or period underwear (17% [95% CI: 14–20] vs 22% [95% CI: 21–23]), and those aged ≥50 were less likely to use regular tampons (32% [95% CI: 28–36] vs. 50% [95% CI: 48–52]). Compared to non-Hispanic White participants, those who self-identified as Asian, Hispanic, or non-Hispanic Black were less likely to use tampons (37% [95% CI: 31–42], 39% [95% CI: 36–43], 31% [95% CI: 27–35] vs. 50% [95% CI: 49–51]); Asian or Hispanic individuals were less likely to use cups (15% [95% CI: 11–18], 15% [95% CI: 13–17] vs. 19% [95% CI: 18–20]). Compared to those with high SES, participants with lower SES used pads more frequently (50% [95% CI: 48–52] vs. 42% [95% CI: 41–44]). Compared to those without any gynecological conditions, those with endometriosis were more likely to use super-absorbent tampons (38% [95% CI: 35–41] vs. 33% [95% CI: 32–35]). Those with longer bleed days used pads more often than those with shorter bleed days; similar patterns were seen for super-absorbent tampons or period underwear (p-values <.05).

Conclusions: In this cohort, tampons and pads were reported with high frequency, in contrast to the low frequency for period underwear/cups. Use differed by sociodemographic, health, and menstrual characteristics. Our findings provide insight into menstrual products use among US adults, informing discussions between clinicians, policymakers, and patients regarding menstrual management.

Tweetable statement:

In a US cohort of 11,455 participants reporting their menstrual product use, tampons and pads were most commonly used, and period underwear or cups were less frequently used. Usage differed by sociodemographic and health characteristics.

Keywords

period product; menstrual hygiene; sanitary pads; tampon; period underwear; menstrual cup; disparities

Introduction

The menstrual cycle is a vital sign—an indicator for health across the lifespan.¹ Using menstrual products is a part of managing menstrual bleeding.^{2,3} Product options, advertising,^{4–6} access, and preferences have shifted over time and place, influenced by personal, social, economic, and historical factors.^{7–13} Within the United States (US), few pioneering studies evaluated product use patterns, including demographic differences.^{14,15} These studies were limited in size or location,¹⁴ and comparisons of newly emerging products (e.g., menstrual cup or period underwear) to historically common products (pads and tampons) were limited.^{16–18}

Taboos, social norms, and generational differences around menstrual products can vary substantially across cultures. While existing studies evaluated differences by race and ethnicity (“Black”/ “Latinx”/ “White”/ “Some other identity”),¹⁵ aggregating as “others” may obscure the underlying differences.^{19,20} Furthermore, period poverty—inadequate access to menstrual hygiene tools like sanitary products, washing facilities, and waste management—is associated with negative health effects globally.^{21,22} Examining variations in menstrual product use across sociodemographic groups could deepen the understanding of usage differences, especially among historically marginalized groups.

Data are also limited on whether product usage differs by gynecological characteristics. Conditions like uterine fibroids and endometriosis could increase physical and financial burdens associated with menstrual products use.^{23,24} Research highlights the importance of considering menstrual product usage in assessing heavy bleeding among patients,²⁵ and the advantages and limitations of estimating flow using traditional products (pad/tampon, including super-absorbent tampon) vs. newer options (cup/period underwear).²⁶ However, data on usage patterns across menstrual characteristics in population studies remains limited.

In this study, we evaluated variations in menstrual product use patterns by sociodemographic, health, and menstrual characteristics, with the goal of describing product usage patterns within a large, demographically diverse US cohort.

Materials and Methods

Study Design and Population

The Apple Women’s Health Study (AWHS) is a prospective digital cohort study. Recruitment began 11/14/2019 and is ongoing. In short, recruitment efforts included a public study website²⁷ with FAQs and background information, social media accounts with recruitment materials posted approximately 1–3 times a week, as well as media events including a podcast,²⁸ media articles, and press interviews after study launch. Further details were described previously.²⁹ Participants are eligible to enroll through the Apple Research app on an iPhone if they live in the United States, are age 18 or older (19 in Alabama and Nebraska, 21 in Puerto Rico), have menstruated at least once, are able to communicate in English, are the sole user of their iCloud account, and provide written informed consent at enrollment.²⁹ This study was approved by the Institutional review Board at Advarra (CIRB #PRO00037562).

We analyzed data from 11,455 female participants enrolled between 11/2019–1/2024 who completed survey questions on self-reported sociodemographic, medical, and menstrual characteristics and menstrual products use. Surveys were distributed at enrollment and updated annually. Using a cross-sectional approach, we evaluated first responses to surveys. Analyses were limited to eligible subsets (Supplemental Figure S1), excluding non-responses or those who selected “I don’t know” or “I prefer not to answer”.

Menstrual Products Use Assessment

The Cycle Tracking Survey asked participants, “What kind of menstrual hygiene products do you typically use? (select all that apply)” with the options of “panty liners”, “sanitary napkins or pads”, “regular tampons”, “super-absorbent tampons”, “a combination of pads or tampons but not used together”, “internal cup menstrual collection device”, “period underwear”, “none”, or “other”. Tampon users were further asked if they typically sleep with a tampon in place at night during their period (“yes” or “no”).

Sociodemographic Characteristics

We evaluated sociodemographic characteristics from self-reported data at baseline: (1) age, categorized as <20, 20–29, 30–39 (referent), 40–49, or 50–90 years old; (2) race and ethnicity, using self-identified categories of “American Indian or Alaska Native”, “Asian”, “Hispanic”, “Middle Eastern or North African”, “Native Hawaiian or Pacific Islander”, “non-Hispanic Black”, “non-Hispanic White” (referent), “none of these fully describe me”, and multiple races (self-identified >1 option); and (3) socioeconomic status (SES): self-perceived score using the MacArthur Scale of Subjective Social Status,³⁰ categorized as 0–3 (low), 4–5 (medium), 6–9 (high, referent).

Gynecological Conditions

We evaluated self-reported gynecological conditions at enrollment. Participants were asked, “Have you ever been diagnosed with any of the following by a doctor or other care provider? (select all that apply)” and the options of “abnormal pap smear”, “endometriosis”, “fibroids”, “infertility”, “polycystic ovary syndrome (PCOS)”, “premenstrual syndrome (PMS) or premenstrual dysphoric disorder (PMDD)”, or “none of the above” (referent).

Menstrual Cycle Characteristics

We assessed self-reported menstrual characteristics: (1) typical menstrual cycle length (MCL): “In the last year, how many days did your typical menstrual cycle last?” with the options of <21, 21–25, 26–31, 32–39, ≥40 days, or period is too unpredictable, (2) typical period days: “In the last year, about how many days did your period typically last, counting only your bleeding days and not spotting days?” with the options of 0, 1, 2, 3, 4, 5, 6, 7, or ≥8 days (categorized to 0, 1–4, 5–7, or ≥8 days).

Statistical Analysis

We summarized the distributions of participant characteristics and reported use for each product, using mean (standard deviation [SD]) and median (interquartile range [IQR]) for continuous variables and numbers (percentages) for categorical variables. To understand

how product usage differs by sociodemographic, health, and menstrual characteristics, we stratified by each covariate. The 95% confidence intervals (CIs) were calculated for the percentages of each product within each group. For covariates with ≥ 3 groups, we evaluated between-group differences using 95% CIs to the referent groups specified above. For variables with mutually exclusive groups, overall differences were evaluated using Chi-square tests.

Sensitivity analyses examined usage variations by self-identified gender, as well as self-identified ancestries among non-Hispanic Asian, Hispanic, and Black participants. We also examined product usage among participants who reported a status of pregnancy, lactation, menopause, or history of hysterectomy/oophorectomy (bilateral). As the question asked of typical product use without a specified timeframe, these individuals may select any options, including “none”.

Data processing and analyses were conducted in Python (version 3.6) and R (version 4.1.2). All statistical tests were two-sided. The results section focuses on reporting the statistically significant differences of relatively large magnitude from the main analysis.

Results

Study sample description

Among the 11,455 participants (Table 1), the median age at enrollment was 33 years (IQR: 26–40). Racial/ethnic distribution was: 0.6% American Indian/Alaska Native, 3% Asian, 7% Hispanic, 0.4% Middle Eastern/North African, 0.1% Native Hawaiian/other Pacific Islander, 5% non-Hispanic Black, 73% non-Hispanic White, and 10% multiple races. Most participants (97%) self-identified as woman. Most participants had at least a college education (81%). Around 40% reported an MCL of 26–31 days, and 45% reported typical bleeding for 5–7 days. Among those who further provided baseline medical history ($n=11,193$), common conditions included abnormal pap smear (28%), PMS/PMDD (14%), PCOS (13%), endometriosis (7%), fibroids (7%), and infertility (5%).

Overall menstrual product use

The most frequently used product was regular tampons (48%), followed by pads (47%), panty liners (43%), super-absorbent tampons (36%), period underwear (20%), internal cups (18%), a combination of pads or tampons but not used together (18%), other products (2%), and 4% reported not using any menstrual product (Figure 1, Supplemental Table S1). Among tampon users, 61% reported sleeping overnight with a tampon. Among the 470 participants reporting no product use (Supplemental Table S2), most were ≥ 40 years or reported unknown/0 typical bleed days, suggesting potentially during perimenopause; some had undergone hysterectomy (28%), oophorectomy (11%), and/or reported being menopausal (8%).

Menstrual product use stratified by sociodemographic factors

The use of tampons, pads, and cups differed by age groups (Figure 2, Supplemental Table S3). Compared to those aged 30–39, younger participants (e.g., 18–19) were more likely

to use pads (57% [95% CI: 53–61] vs. 45% [95% CI: 43–46]) and less likely to use cups (12% [95% CI: 10–15] vs. 21% [95% CI: 19–22]) or period underwear (17% [95% CI: 14–20] vs 22% [95% CI: 21–23]). Those aged ≥ 50 were less likely to use regular tampons (32% [95% CI: 28–36] vs. 50% [95% CI: 48–52]) or cups (7% [95% CI: 5–10] vs. 21% [95% CI: 19–22]). The use of products differed by self-identified race and ethnicity (Figure 3, Supplemental Table S4). Compared to non-Hispanic White participants, those who self-identified as non-Hispanic Asian, Hispanic, or non-Hispanic Black were less likely to use tampons (e.g., regular tampons: 37% [95% CI: 31–42], 39% [95% CI: 36–43], 31% [95% CI: 27–35] vs. 50% [95% CI: 49–51]); non-Hispanic Asian or Hispanic participants were less likely to use cups (15% [95% CI: 11–18], 15% [95% CI: 13–17] vs. 19% [95% CI: 18–20]). The use of all products (except regular tampons) differed by subjective SES (Figure 4, Supplemental Table S5). Compared to the high SES group, those reporting low SES tend to use pads more (50% [95% CI: 48–52] vs. 42% [95% CI: 41–44]), but less use of cups (17% [95% CI: 15–18] vs. 21% [95% CI: 20–23]) or period underwear (19% [95% CI: 18–20] vs. 22% [95% CI: 20–23]).

Menstrual product use stratified by gynecological health

Compared to those without gynecological conditions (Figure 5, Supplemental Table S6), those with infertility had a higher prevalence of using pads, super-absorbent tampons, or period underwear; those with endometriosis reported using super-absorbent tampons more (38% [95% CI: 35–41] vs. 33% [95% CI: 32–35]); those with fibroids reported sleeping overnight with tampons more (69% [95% CI: 64–73] vs. 60% [95% CI: 59–62]); differences of smaller magnitudes were observed for participants with PCOS, abnormal pap smear, or PMS/PMDD.

Menstrual product use stratified by menstrual cycle characteristics

Compared to those with MCL of 26–31 days, those with shorter (e.g., < 21 days) or longer MCL (≥ 40 days or too irregular) used cups less (< 21 days: 10% [95% CI: 8–12], ≥ 40 days: 18% [95% CI: 14–21], too irregular: 15% [95% CI: 13–16], compared to 26–31 days: 23% [95% CI: 21–24]; Figure 6, Supplemental Table S7). There were incremental increases per category of product (pads, super-absorbent tampons, cups, or period underwear) as the number of typical bleed days increased (Figure 7, Supplemental Table S8).

Sensitivity analyses

Participants self-identifying as gender queer/non-binary or man (3.1%) reported much lower use of tampons or cups compared to self-identified women (Supplemental Figure S2). Most Asian ancestry groups, except Korean, had lower tampons usage than non-Hispanic White participants, and participants of Indian ancestry reported the lowest usage (Supplemental Figure S3). Participants who identified Cuban ancestry reported lower usage of cups (Supplemental Figure S4). Among non-Hispanic Black participants (Supplemental Figure S5), 27 participants who identified Haitian ancestry had the lowest usage of regular tampons and very low usage of cups. Individuals who reported menopausal or had undergone hysterectomy often reported no product use, but 10–20% still reported using panty liners/pads (Supplemental Figures S6 and S7).

Comment

Principal Findings

In this large US cohort, tampons and pads were most commonly used, and period underwear or cups were less commonly used. Non-Hispanic Asian, Hispanic, or non-Hispanic Black participants reported lower usage of tampons or cups. Participants with low SES reported higher usage of pads and lower usage of cups or period underwear. Usage also differed by pre-existing gynecological conditions and menstrual cycle characteristics. For example, those with endometriosis or >7 bleeding days used more super-absorbent tampons, while those whose MCL deviated from 21–39 days reported lower cups or period underwear use.

Result in the Context of What is Known

Few US studies have evaluated menstrual product usage. One study (1980s) of 1,601 clinic patients reported the majority using pads only, followed by both tampons and pads, then tampons only.³¹ A recent study in California (n=357) reported pads as most common (76%) followed by tampons (52%) and menstrual cups (11%).¹⁴ Another study in 2023 (in California and Washington D.C.) found unscented pads as the most commonly used (70%), followed by unscented tampons (47%), menstrual cups (11%), scented pads (10%), and scented tampons (5%).¹⁵ We similarly found pads being commonly used but with lower percentages, possibly due to distinguishing panty liners from pads. We also assessed newly emerging products like period underwear and evaluated regular versus super-absorbent tampons. However, we lacked data on products fragrance. Existing studies found either lower tampons or cups use among Black or Hispanic individuals compared to White individuals, or no differences, with individuals of older age less likely to use tampons or cups.^{14,15,31} Our age-stratified findings align with these studies, and our analyses using non-aggregated self-identified race and ethnicity groups provided more information. Notably, the low use of tampons or cups among those self-identifying as non-Hispanic Asian (especially Indian ancestry), Hispanic, Middle Eastern/North African, non-Hispanic Black (especially Haitian ancestry), or America Indian/Alaska Native, have rarely been reported before. In some cultures, tampon use has been discouraged due to concerns of impeded blood flow or representing a loss of virginity,^{7,32–37} and traditional practices around menstrual hygiene may continue in families with young women who migrate to another country such as the United States, likely driven by cultural norms and socioeconomic status.^{38,39} Lower reporting of tampons/cups for those who self-identified as gender queer/non-binary/man is also in line with literature on queer menstruation and non-penetrative menstrual product use.⁴⁰ Our usage findings of variation by SES, gynecological conditions, and menstrual cycle characteristics and the distributions of a behavioral factor (sleeping with tampon) have not previously been reported.

Clinical Implications

Our study evaluating eight different types of menstrual products in a large study of 11,455 participants across the US may help anchor and contextualize the menstrual product use patterns for researchers, policy makers, healthcare providers, and the general public. Our findings on product use variation by SES provides nation-wide data for policy makers trying to pass menstrual hygiene laws, which multiple states have done, such as the

Menstrual Equity for All Act initiated in 2021 which would provide free menstrual products in public spaces.⁴¹ The studied usage variations by age could also serve as reference for deepening menstrual products options conversations among individuals of various reproductive stages including the menopausal transition.^{1,42} The variations by race/ethnicity, gender, and SES we describe may provide more context when discussing menstrual health issues with patients of historically marginalized groups in the United States.^{43–45} Variations by gynecologic conditions and menstrual cycle characteristics appeared consistent with clinical manifestations of these conditions (e.g., more super-absorbent tampon use among those with endometriosis,⁴⁶ products usage increased as number of typical bleeding days increase). These findings may provide a useful reference to encourage patients with health conditions to make informed choices regarding menstrual product options.³ Moreover, these data provide insights into how menstruating individuals cope with potentially excessive/prolonged flow (e.g., those with certain gynecologic conditions or those with >7 bleeding days) through using more products with enhanced absorbency. This highlights the importance of considering product usage when evaluating heavy menstrual bleeding.^{25,26,47}

Research Implications

Despite the gaps our work has filled in documenting product use patterns and variations, there remain unanswered questions like product access and whether there is any health implications related to the usage of the varied menstrual products.^{15,48,49} Previous literature identified age, race, educational level, income, and family size as predictors of menstrual product insecurity. Menstrual hygiene is a public health concern as those experiencing period poverty, which has increased in recent years,⁵⁰ may turn to unsanitary alternatives.⁵¹ Understanding population-level usage patterns must be interpreted in the context of cost (particularly given the recent price increases of approximately 36–41% since 2019)⁵² and emerging reusable products that may be less expensive overall, but have a large up-front cost. Future research including cost and accessibility analyses is needed to develop solutions to address period poverty. We found that 20% of participants used period underwear. Recent studies have reported the presence of endocrine disrupting chemicals in some period underwear products,^{53,54} although the potential health impacts remain unclear. Given the sizeable proportion of participants using period underwear in this population, future research to validate the presence of chemicals, understand absorption, and investigate health implications are warranted.

Strengths and Limitations

Limitations include a one-time self-report of typical product usage. Potential misclassification may have also arisen depending on participants' interpretation of each provided option. We did not collect data on nativity/acclimation, religious beliefs, or other factors that may co-vary with ethnicity, thus were unable to directly investigate these as contributors to the ethnicity variation we observed. We did not have brands, ingredients, or fragrance information or barriers to access, stigma, or affordability information relevant to period poverty; however, we were able to observe trends through self-reported sociodemographic characteristics. Future research is needed to evaluate chemicals in menstrual products and disparities of access. Lack of detailed contraceptive usage data (brand, ingredient, and dosage) also prevented us from accounting for the impact of certain

hormonal contraceptives on product usage patterns, particularly for participants experiencing contraceptive-induced amenorrhea. Lastly, our results may not be generalizable to all US individuals who menstruate. Potential selection bias may occur due to self-selection into the study, and the study is limited to Apple iPhone users with concerns of generalizability to other populations using other digital platforms.

This study has unique strengths. First, we included a comprehensive evaluation of eight types of products from a large population across the US, allowing for sufficient statistical power to detect differences by sociodemographic factors, even for previously understudied groups. Second, to the best of our knowledge, this study is the first to quantitatively assess and report differences in menstrual product use by gynecological/menstrual characteristics, providing population-based reference data. Third, we assessed another behavioral aspect of product use—sleeping with tampons—which may inform future research on associated risks and perceptions. Overall, determining menstrual products usage patterns represents a critical step towards recognizing the menstrual cycles as a vital sign.

Conclusions

In a large US cohort, tampon or pads were most used, and period underwear and cups were used by individuals of certain demographics. Usage varied by sociodemographic (race, ethnicity, ancestry, gender identity, and SES), gynecological conditions, and menstrual characteristics. While further research on access, affordability, and health implications are needed, our findings serve in guiding consultations around menstrual cycle management in clinical and policy settings, particularly when facing patients from diverse backgrounds and with varying health conditions, ultimately enhancing the quality of life for menstruating individuals.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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This study is not a clinical trial. However, the parent cohort, Apple Women's Health Study, is registered on [Clinicaltrials.gov](https://clinicaltrials.gov) (NCT04196595): <https://classic.clinicaltrials.gov/ct2/show/NCT04196595>

Data Sharing Statement

Aggregated data that support the findings of this study may be available upon reasonable request from the corresponding author & senior author. Any request for data will be evaluated and responded to in a manner consistent with policies intended to protect participant confidentiality and language in the Study protocol and informed consent form.

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AJOG at a Glance:**Why was this study conducted?**

Few small US-based studies from specific regions found differences in menstrual product use across age or aggregated race and ethnicity. We evaluated menstrual product use and variations in a large US cohort of diverse sociodemographic, health, and menstrual characteristics.

What are the key findings?

Tampons/pads were most commonly used, and period underwear/cups were less commonly used. Hispanic or non-Hispanic Asian/Black participants reported lower usage of tampons/cups. Participants with low socioeconomic status used cups/period underwear less. Those with endometriosis used more super-absorbent tampons; those with irregular menstrual cycles used less cups/period underwear.

What does this study add to what is already known?

Our findings contextualize menstrual product use in the US, which could be useful for discussions between clinicians, policymakers, and patients regarding menstrual management.

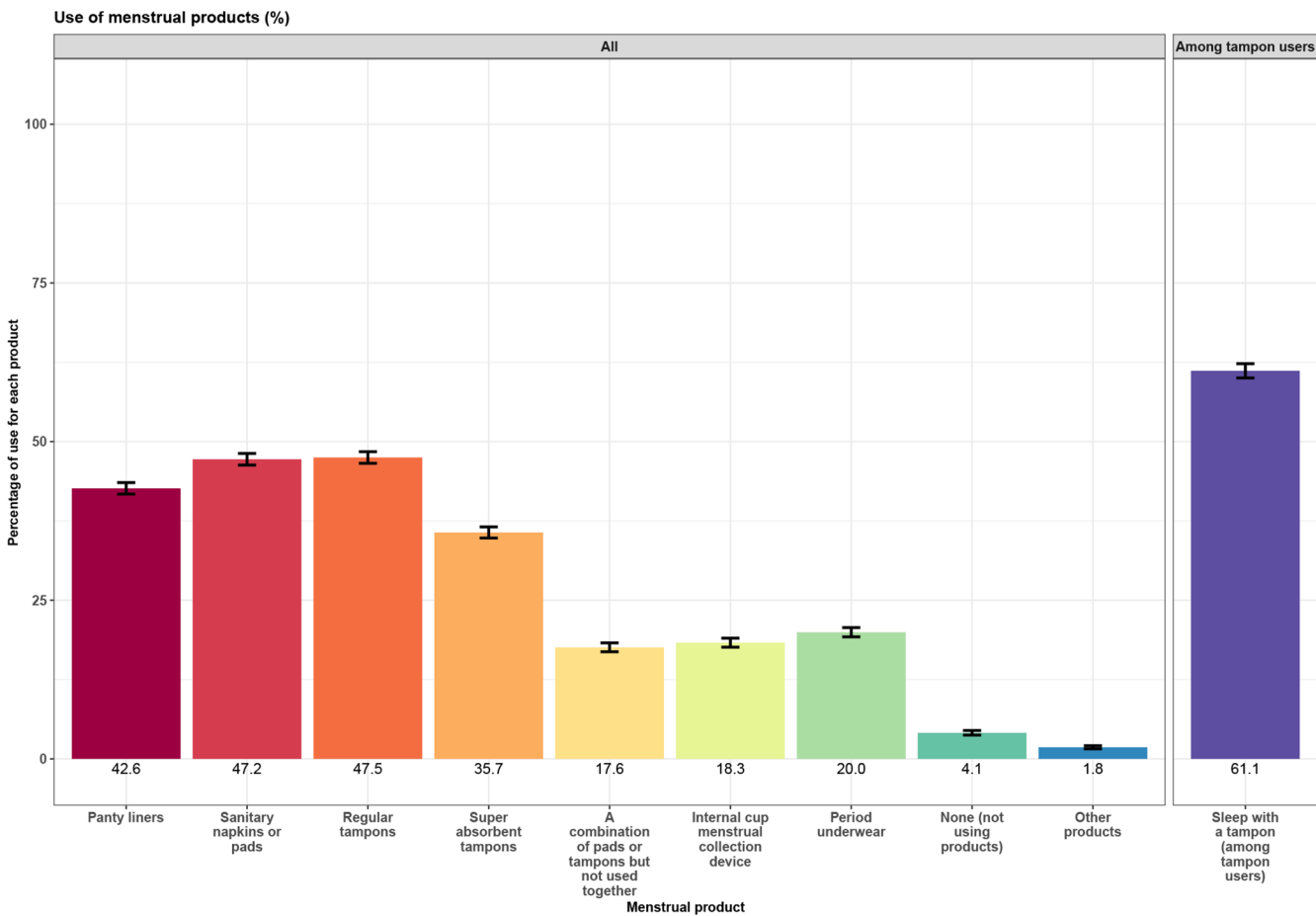


Figure 1. Menstrual product use among the 11,455 participants. Each colored bar represents the percentage of reported use for each type of menstrual product (with numeric values of the percentages also labelled at the bottom of each bar). Error bars represent the 95% confidence intervals of the percentages.

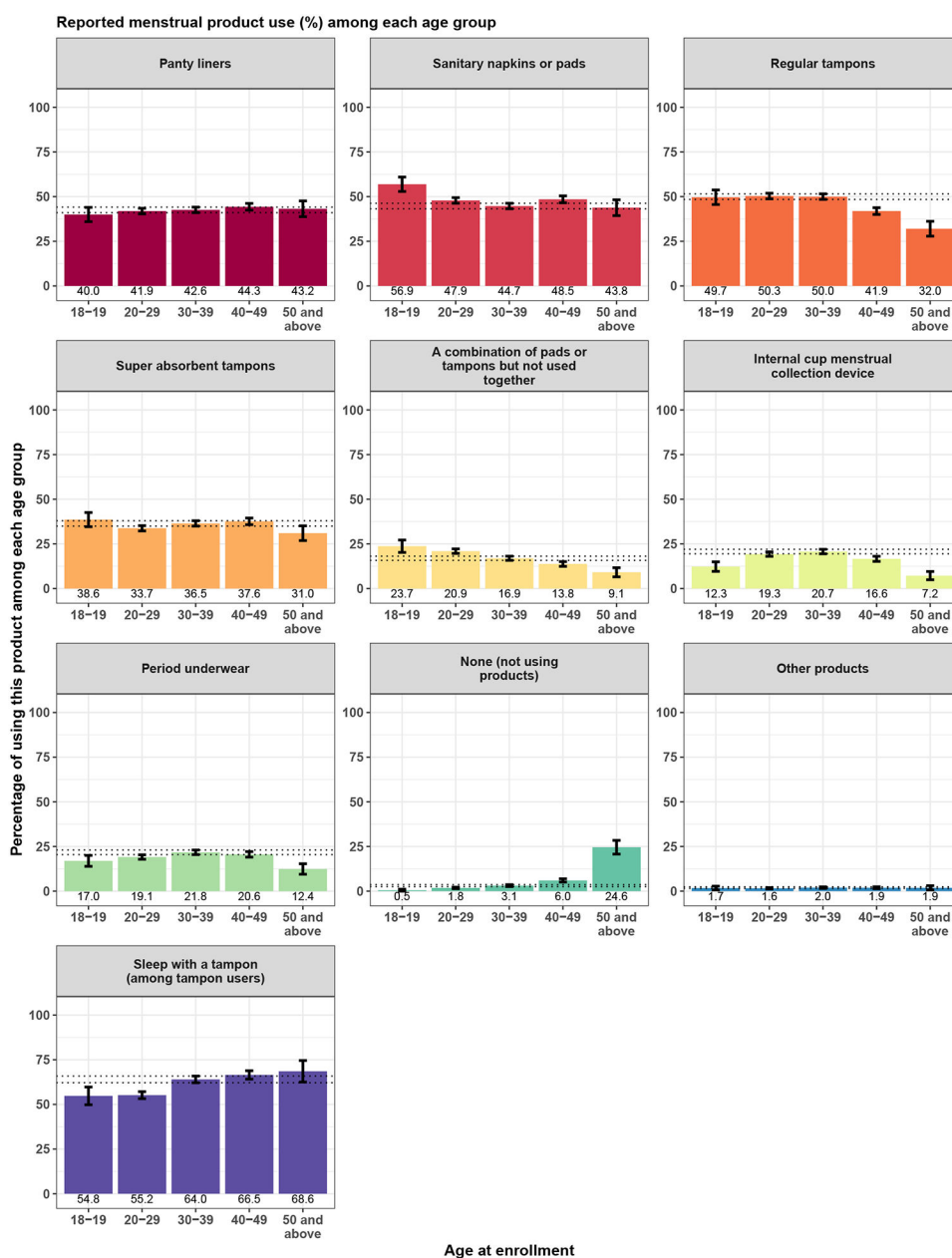
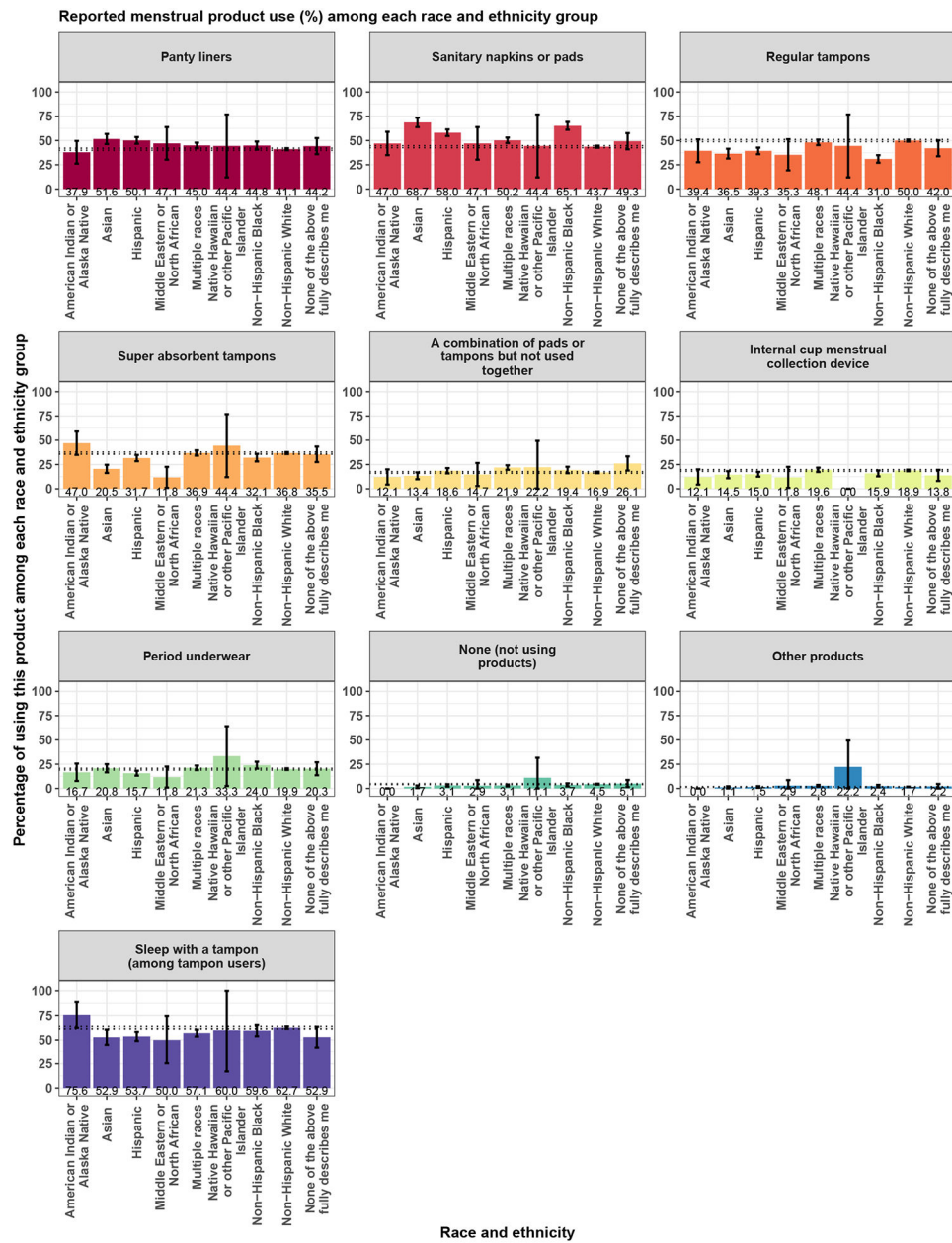


Figure 2.

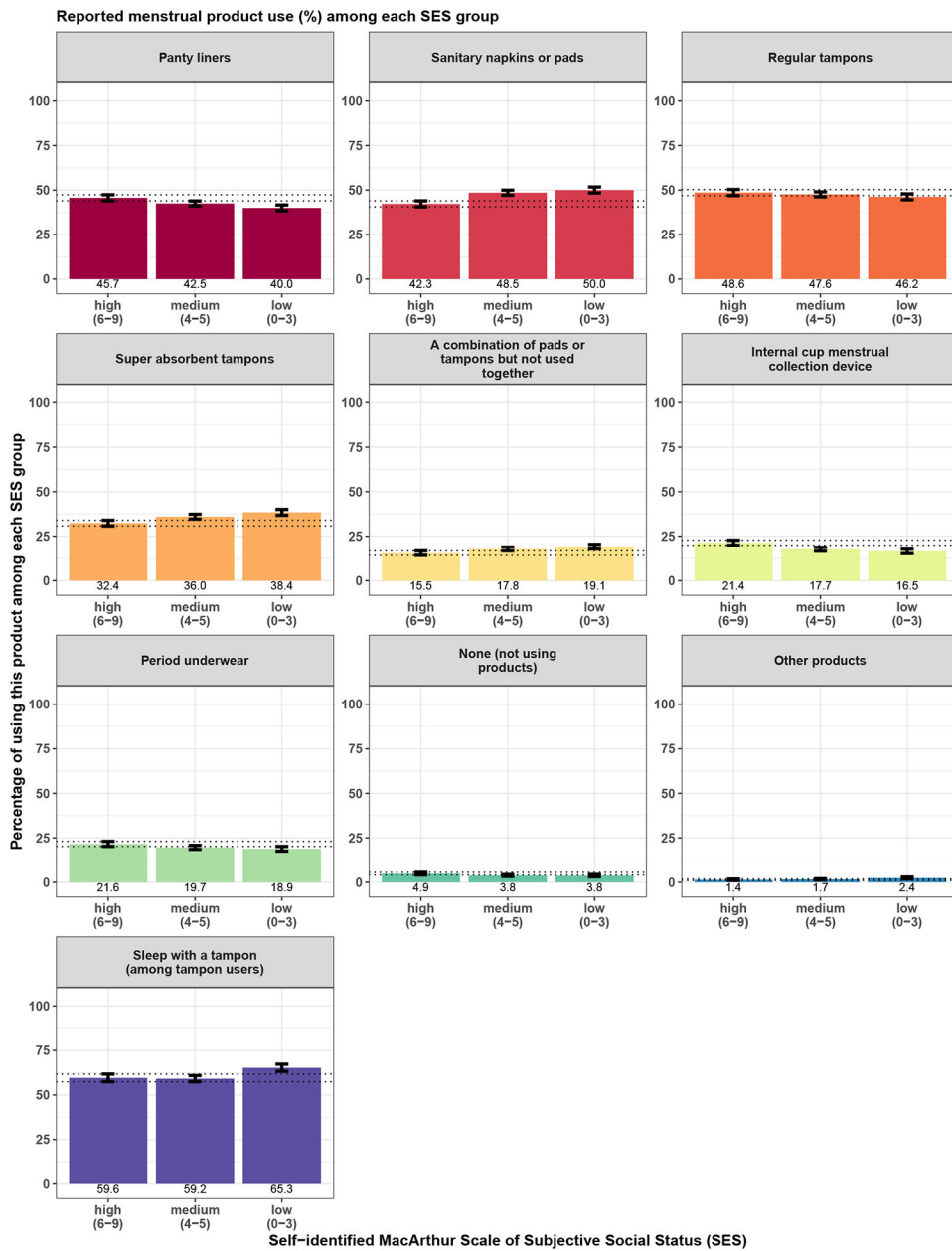
Menstrual product use patterns by age at enrollment groups.

Each colored bar represents the reported frequency of use for a type of menstrual product among individuals in each age group (with numeric values of the percentages also labelled at the bottom of each bar). Error bars represent the 95% confidence intervals (CIs) of the percentages. Dotted lines represent the 95% CIs of the age 30-39 group.

**Figure 3.**

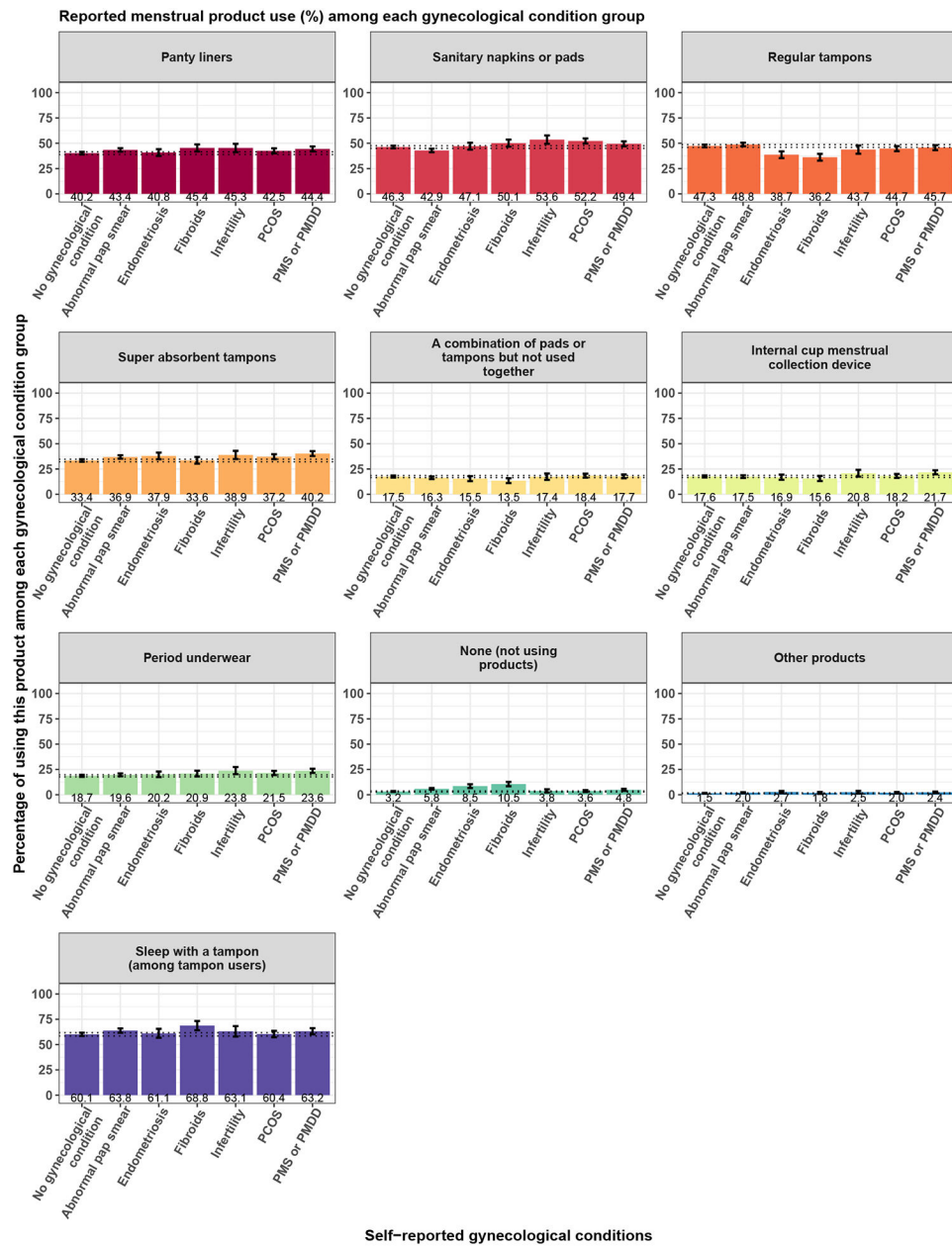
Menstrual product use patterns by self-identified race and ethnicity groups.

Each colored bar represents the reported frequency of use for a type of menstrual product among individuals in each self-identified race and ethnicity group (with numeric values of the percentages also labelled at the bottom of each bar). Error bars represent the 95% confidence intervals (CIs) of the percentages. Dotted lines represent the 95% CIs of the non-Hispanic White group.

**Figure 4.**

Menstrual product use patterns by self-identified socioeconomic (SES) groups.

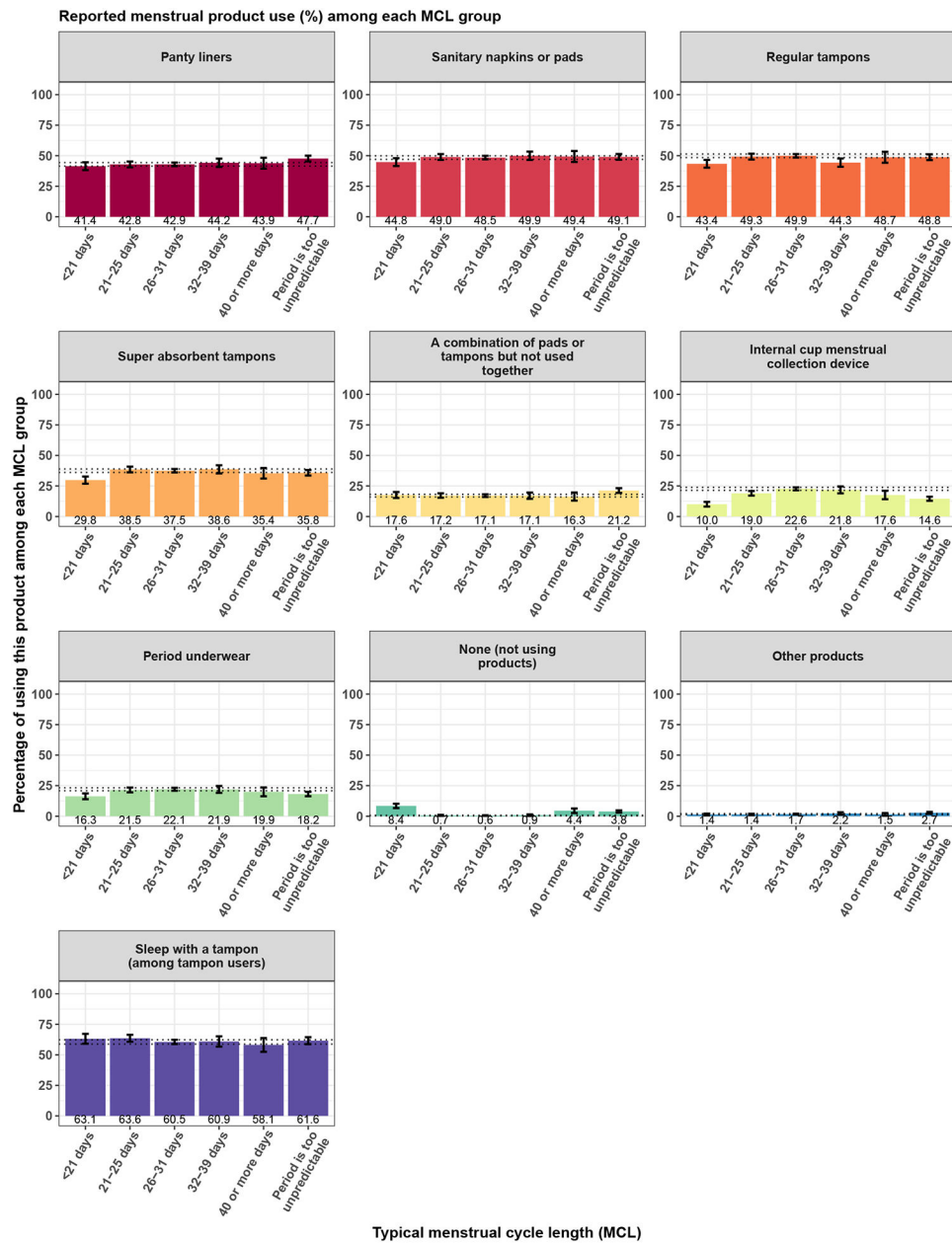
Each colored bar represents the reported frequency of use for a type of menstrual product among individuals in each self-identified SES group (with numeric values of the percentages also labelled at the bottom of each bar). Error bars represent the 95% confidence intervals (CIs) of the percentages. Dotted lines represent the 95% CIs of the high SES group.

**Figure 5.**

Menstrual product use patterns by self-reported gynecological condition groups.

Abbreviations: PCOS, polycystic ovary syndrome; PMS, Premenstrual syndrome; PMDD, Premenstrual dysphoric disorder.

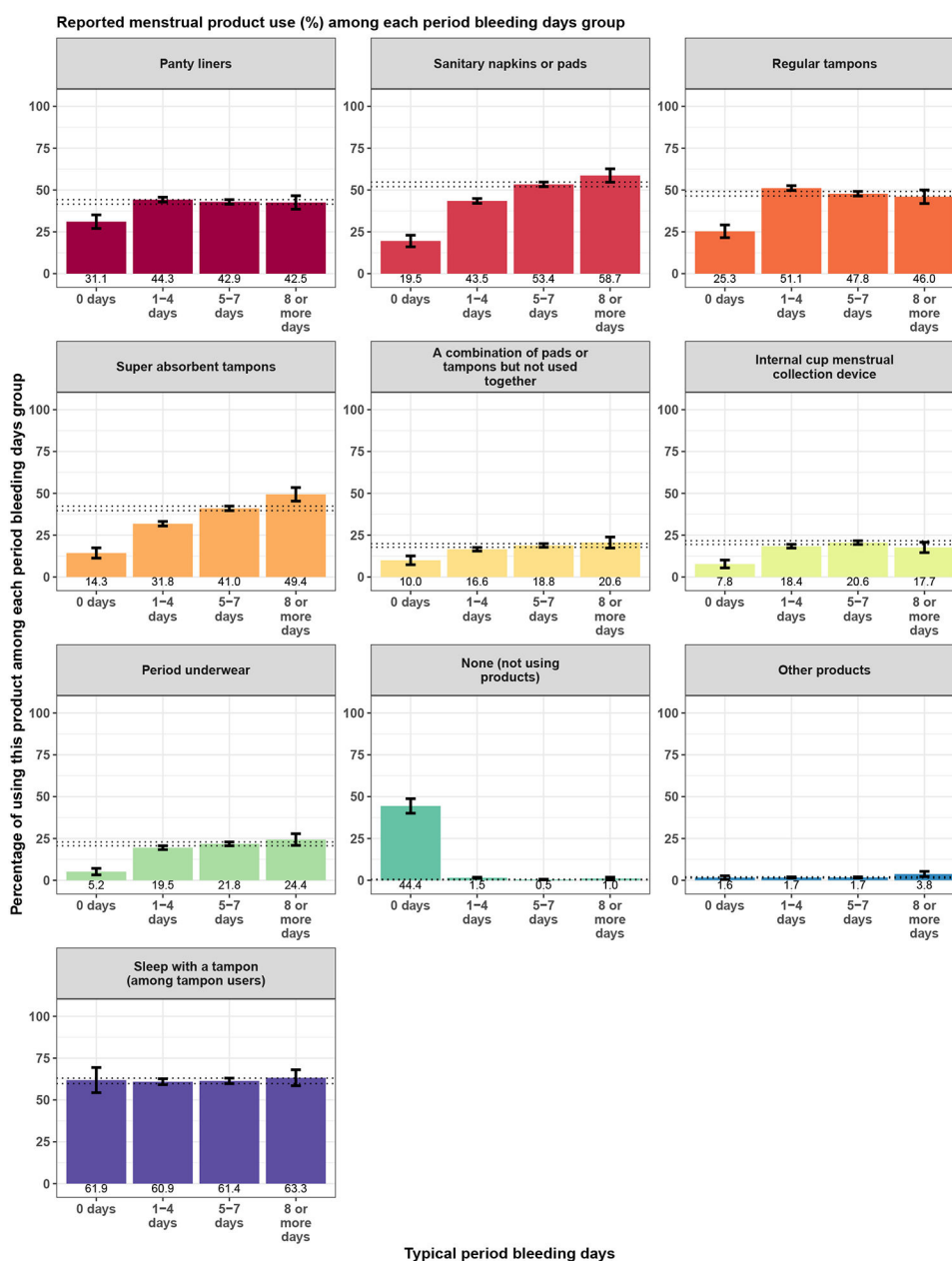
Each colored bar represents the reported frequency of use for a type of menstrual product among individuals in each self-reported gynecologic condition group (with numeric values of the percentages also labelled at the bottom of each bar). Error bars represent the 95% confidence intervals (CIs) of the percentages. Dotted lines represent the 95% CIs of the group without any of the listed conditions.

**Figure 6.**

Menstrual product use patterns by self-reported typical menstrual cycle length.

Abbreviations: MCL, menstrual cycle length.

Each colored bar represents the reported frequency of use for a type of menstrual product among individuals in each self-reported MCL group (with numeric values of the percentages also labelled at the bottom of each bar). Error bars represent the 95% confidence intervals (CIs) of the percentages. Dotted lines represent the 95% CIs of the group with typical MCL of 26–31 days.

**Figure 7.**

Menstrual product use patterns by self-reported period bleeding days.

Each colored bar represents the reported frequency of use for a type of menstrual product among individuals in each self-reported period bleeding days group (with numeric values of the percentages also labelled at the bottom of each bar). Error bars represent the 95% confidence intervals (CIs) of the percentages. Dotted lines represent the 95% CIs of the group with typical period bleeding days of 5–7 days.

Table 1.

Baseline sociodemographic and health characteristics of 11,455 participants from AWHs.

Characteristics, based on self-report at baseline ^a	All (N = 11,455)
Age at enrollment, years	
Mean ± SD	33.2 ± 9.5
Median (IQR)	33 (26 – 40)
Age at enrollment, n (%)	
18–19	578 (5.0)
20–29	3859 (33.7)
30–39	3931 (34.3)
40–49	2603 (22.7)
50 and above	484 (4.2)
Race and ethnicity, n (%)	
American Indian or Alaska Native	66 (0.6)
Asian	351 (3.1)
Hispanic	796 (6.9)
Middle Eastern or North African	34 (0.4)
Native Hawaiian or other Pacific Islander	9 (0.1)
Non-Hispanic Black	542 (4.7)
Non-Hispanic White	8300 (72.5)
None of the above fully describes the participant	138 (1.2)
Multiple races ^b	1181 (10.3)
Prefer not to answer	38 (0.3)
Subjective SES scale, n (%)	
High (6–9)	3230 (28.2)
Medium (4–5)	4815 (42.0)
Low (0–3)	3385 (29.6)
Self-identified as Woman, ^c n (%)	11087 (96.8)
Self-identified as Man, ^c n (%)	16 (0.1)
Self-identified as Trans Woman, ^c n (%)	3 (0.0)
Self-identified as Trans Man, ^c n (%)	34 (0.3)
Self-identified as Gender Queer or Non-Binary, ^c n (%)	371 (3.2)
Self-identified another gender identity, ^c n (%)	33 (0.3)
Prefer not to report gender identity, ^c n (%)	37 (0.3)
Education level, n (%)	
High school or less	2154 (18.8)
Some college or college	6855 (59.8)
Graduate school	2390 (20.9)

Characteristics, based on self-report at baseline ^a	All (N = 11,455)
Prefer not to answer	56 (0.5)
Typical menstrual cycle length, n (%)	
<21 days	927 (8.1)
21–25 days	1671 (14.6)
26–31 days	4521 (39.5)
32–39 days	821 (7.2)
40 or more days	472 (4.1)
Period is too unpredictable	1718 (15.0)
Don't know	1154 (10.1)
Prefer not to answer	171 (1.5)
Typical period (bleeding) days, n (%)	
0 day	502 (4.4)
1–4 days	4576 (39.9)
5–7 days	5093 (44.5)
8 or more days	583 (5.1)
Don't know	544 (4.7)
Prefer not to answer	157 (1.4)
Among 11,193 participants who further filled out a baseline Medical History Survey:	
Reported having an abnormal pap smear, ^d n (%)	3131 (28.0)
Reported having endometriosis, ^d n (%)	828 (7.4)
Reported having fibroids, ^d n (%)	780 (7.0)
Reported having infertility, ^d n (%)	558 (5.0)
Reported having PCOS, ^d n (%)	1499 (13.4)
Reported having PMS or PMDD, ^d n (%)	1588 (14.2)
Reported none of the above gynecological conditions, ^d n (%)	5524 (49.4)

Abbreviations: AWHs, Apple Women's Health Study; SD: standard deviation; IQR: interquartile range; SES, socioeconomic status; PCOS, polycystic ovary syndrome; PMS, Premenstrual syndrome; PMDD, Premenstrual dysphoric disorder.

^aNumbers may not add to total N due to non-response.

^bMultiple races correspond to those who selected more than one race and ethnicity category.

^cParticipants were given the option to select more than one gender identity. This data reflects precisely participant's self-identified gender, which may sometimes be inconsistent with the exclusion criteria based on reported sex assigned at birth.

^dParticipants were given the option to select more than one gynecological condition(s) from the list in the survey question "Have you ever been diagnosed with any of the following by a doctor or other care provider? (select all that apply)". "None of the above" is an exclusive choice.